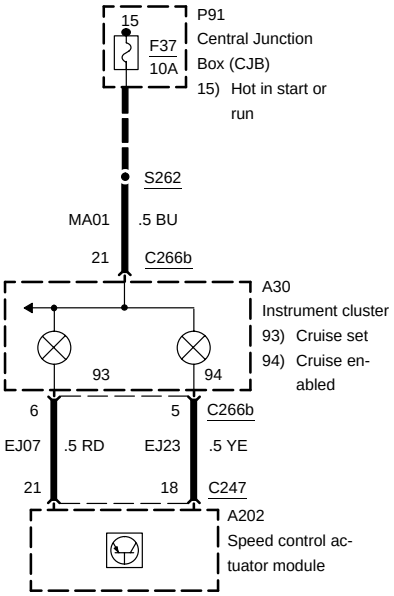conventional cluster -5